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A preliminary assessment of the performance of DeFi cryptocurrencies in relation to other financial assets, volatility, and user-generated content

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ABSTRACT

After the so-called "crypto-winter", decentralised finance (DeFi) is reviving interest in cryptocurrency amongst the scientific community, public and private institutions, and investors. DeFi is a novel disruptive process that promotes the use of blockchain technology for creating and issuing all kinds of financial products and services. This study aimed to measure the relationship amongst the returns of DeFi tokens, other traditional assets, and user-generated content. While the relationship between other crypto assets and traditional assets has been researched, this has not been done on DeFi assets. This study uses a logit-probit model over a database comprising the daily returns of 13 DeFi, VIX, S&P GSCI Crude Oil Index, and S&P GSCI Gold Index, and the daily variation in DeFi mentions in Telegram chats and Twitter. The results show that all variables except the S&P GSCI crude oil index returns and the daily variation in Twitter mentions were significant. This suggests that DeFi acts, similar to other crypto assets, as a safe haven. This study contributes to the literature on decentralised finance tokens as investment assets, which requires much more research.

1. Introduction

The financial crisis of 2007 had an unexpected effect; an unknown hacker published a paper (Nakamoto, 2008) describing an unsupervised digital currency with the aim of avoiding the need for a central authority. After this paper (Nakamoto, 2008), Satoshi released an open-source software that started Bitcoin as a decentralised digital cryptocurrency. The lack of central governance is a key feature that attracts interest and at the same time raises concerns about its risks (Grant and Hogan, 2015). The cost of avoiding a central node increased when Bitcoin began to increase in popularity because of electricity consumption (Fairley, 2017). The increase in transaction fees to keep mining profitable makes it unusable as a payment system for small amounts of money. Satoshi's paper talks about "electronic cash" to make "electronic payments", but in reality after years of being in use, Bitcoin is mainly used as an asset (Baur et al., 2018), either comparable with gold (Dyhrberg, 2016) or not (Klein et al., 2018). Over the years, Bitcoin has gained attention and market share. Another hacker published a white paper describing a blockchain of smart contracts called Ethereum (Buterin et al., 2014), which is a disruptive technology that can act as a digital, immutable, and programmable public agreement. This new technology has huge implications in mitigating informational asymmetry, enhancing competition, and improving consumer welfare; yet, it may also lead to collusion (Cong and He, 2019).

The appearance of altcoins provided an alternative investment opportunity. A good and popular example in Ethereum is an initial coin offering (ICO). An ICO acts as an initial public offering (IPO) of cryptocurrencies worldwide. It serves as an early-stage crowdfunding method to raise funds for a new cryptocurrency before they are publicly traded (Gans and Catalini, 2018).

Ethereum released the ERC20 feature, adding the possibility of creating new cryptocurrencies easily over the Ethereum blockchain. This led to an explosion of ICOs and signs of a bubble, but also great returns in compensation for the assumed risks (Benedetti and Kostovetsky, 2021). This generated great volatility in the stock market (Akyildirim et al., 2020). ICOs grew exponentially from 2015 to the second quarter of 2018. The highest amount of USD 6 billion was raised in the first quarter of 2018. After that, sales dropped remarkably,

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coinciding with the development of law enforcement actions in many countries worldwide. Thus, the exponential growth of ICOs accompanied by the rise in problems related to security and investor protection led to increasingly concerned reactions by public and private regulators (Domingo et al., 2020). The Securities and Exchange Commission (SEC) started to consider tokens as a possible asset under securities regulation, in practice banning US citizens from investing (Hacker and Thomale, 2018). Security token offering (STO) appeared as an alternative to ICOs, because it complies with regional regulators, but the amount of capital raised by STOs is far lower than that of ICOs. Almost simultaneously, in the beginning of 2018, Bitcoin suffered depreciation. The depreciation of the value of Bitcoin, depreciation of ETH, and demise of the ICO market led to the so-called crypto winter.

Finance professionals who used ICOs to develop Ethereum-based financial products led the emergence of the DeFi market in the beginning of 2020. DeFi financial products built into Ethereum smart contracts play a significant role in the development of the Ethereum network. It is believed that they will likely continue to play an increasingly significant role in the future (Wulf, 2020).

The purpose of this research is to study the relationship between DeFi tokens, traditional assets, and user-generated content. DeFi is a very new technology, and very few studies based on volatility have been conducted. Therefore, this is a preliminary study of this new field. We use a dataset comprising the daily returns of 13 DeFi, VIX, S&P GSCI Crude Oil Index, S&P GSCI Gold Index, and the daily variation of DeFi mentions in Telegram chats and Twitter, over the study period covering 1 July 2018–27 July 2020. After thoroughly describing the decentralised finance from different perspectives and angles in relation to traditional finance, this study aims to analyse the influence of the variables mentioned on DeFi returns. This study contributes to the advancement of knowledge on this novel topic by examining the impact of selected sentiment variables, such as Telegram chats and Twitter mentions, on DeFi returns. The results of this study can guide policymakers as well as public and private investors in their decisions towards more efficient markets.

This paper is structured as follows: Section 2 provides the theoretical framework and formulates the hypotheses; Section 3 describes the method, including the sample, variables, and model; Section 4 presents the main empirical results; Section 5 discusses the results; Section 6 presents the implications; and Section 7 provides the conclusion and discusses future work.

2. Literature review

2.1. DeFi

From a broad perspective, DeFi refers to a recent disruptive process that promotes the use of blockchain technology, smart contracts, decentralised ledgers, and applications based on open protocols or modular frameworks for creating and issuing all kinds of financial products and services. Ethereum and other smart contract platforms allow any code to be run using blockchain technology. This opens the possibility of creating decentralised applications (Dapps), decentralised autonomous organisationsDecentralized Autonomous Organizations (DAOs), and decentralised financial services (DeFi). The financial and technical complexity is much higher than that of the ICOs based on ERC20, but the efforts, funding, and interest in the crypto community during these years seem to open a new and exciting possibility for decentralised financial services.

The term decentralised finance is very wide because most blockchain projects are decentralised by definition, and any cryptocurrency (e.g. Bitcoin) can be categorised as a financial asset (Baur et al., 2018). In all its nuances, most of the literature considers DeFi as the replication of financial services over blockchain. Some authors provide a more precise categorisation of the services (Gudgeon et al., 2020; Schär, 2020; Katona 2021), while others have a broader focus (Chen and Bellavitis, 2020). Still, some others focus on specific interoperability services like oracles (Liu et al., 2020). Undoubtedly, the amount of money locked in DeFi smart contracts is growing exponentially, from USD 1 billion in June 2020, to USD 2 billion in July 2020, and to USD 4 billion in August 2020. The tradeable tokens behind the DeFi projects surpassed USD 10 billion in market capitalisation. In other words, the use of these new services seems to be exploding, together with the market value of organisations related to decentralised finance.

2.2. Decentralised financial services

Lending and borrowing are services defined by a smart contract asking for, or offering, cryptocurrencies paying or earning an interest rate, respectively. Platforms such as Aave, Compound, Yearn finance, Just, or Kava offer these services in the form of loans or flash loans. Loans require borrowers to over guarantee their loans, allowing the lender to redeem pledged collateral in the event of default by the borrower. Flash loans allow borrowing without a guarantee because the atomicity of the smart contract protects the loaned amount. Therefore, a loan not repaid with interest means that the whole transaction will be reversed (Gudgeon et al., 2020; Schär, 2020). Flash loans are not easily comparable to traditional services because they rely on the power of smart contracts. For example, if someone with a small amount of money wants to buy a house and sell it immediately in the same operation, they could use a flash loan, avoiding the need to have all the money required by the operation. Although they are becoming popular for arbitrage operations, they have safety problems (Qin et al., 2020) because of attacks related to oracle manipulation and the pump and arbitrage scheme.

Stablecoins are crypto assets with a price-stabilization mechanism pegged to cryptocurrency, fiat money, or exchange-traded commodities. USDT and USDC are cryptocurrencies paired with a dollar but centralised, as each USDT or USDC is backed by a USD. When stablecoins are backed by another cryptocurrency and regulated by smart contracts, they are considered decentralised or DeFi. The most popular stablecoin is DAI. One DAI is equated to one USD and uses a stabilisation mechanism to protect against volatility. In fact, to borrow DAIs, a user must pledge an over-guaranteed number of cryptocurrencies. When the price of DAI differs from that of its peg, arbitrageurs are incentivised to buy or sell DAI to compensate for supply and demand movements. If the collateral cryptocurrency devaluates in relation to the USD, the borrower must increase the amount of collateral to keep the ratio above some liquidation threshold; otherwise, the borrow position will be liquidated at a discount, and a penalty fee will be charged (Qin et al., 2020).

Exchanges are key components of blockchain ecosystems. They allow changes between fiat and cryptocurrency, and are the main entrance of cryptocurrencies. Some of them offer more products, such as staking, options, or derivatives, and they have initial exchange offerings (IEOs), which is their own version of ICOs. IEOs are centralised, and their efficiency is based on trust in the exchange operator, because to trade in them, traders must deposit assets in the exchange account, which can be accessed by the centralised exchanges. DeFi proposed decentralised exchange (DEX) as an alternative to centralised exchanges. In DEX, users maintain control of their assets instead of the exchange account. Trade execution occurs atomically through a smart contract in which both parties trade around one indivisible transaction (Schär, 2020). This mode of operation removes intermediaries and adds transparency, but it also demands interoperability with other services such as oracles or lending to have liquidity. Balancer, Uniswap, 0x, Kiber network, Bancor, and Loopring are examples of actual decentralised exchanges.

Decentralised derivatives are tokens that derive their value from the performance of an underlying asset, the result of an event, or the behaviour of any other observable variable (Schär, 2020). Synthetix is a derivative liquidity protocol on Ethereum which allows for the issuance and trading of synthetic assets. Each synthetic asset (SYNTH) is an

ERC20 token that replicates external asset prices and uses smart contract infrastructure underpinned by the Synthetix Network Token (SNX) as collateral. Interested parties are rewarded for their support of the system with a proportional share of the fees generated by the activity. Consequently, the SNX value is directly related to the use of the network it collateralises.

2.3. Oracles

Oracles are systems that provide trustworthy information from the real world (off-chain) to blockchain (on-chain). They are not services for the end-user, but are crucial for the DeFi ecosystem. Oracles provide real-time information about assets used as collateral and redemptions, such as market prices. As price is very volatile, the security of many DeFi systems relies on oracles. An oracle introduces data into a smart contract from a third-party source, assuring veracity in different ways (Liu et al., 2020). The most successful project in this area was Chainlink (LINK). It raised money from an ICO in 2017 at a price of USD 0.11 per token, and at the time this paper was written (August, 2020), the price of each token was 150 times higher (USD 16.87).

2.4. DeFi objectives and challenges

The main objective of decentralised finance is to provide financial services over the blockchain in a decentralised manner. As DeFi originated from blockchain systems, it seems that everything is already decentralised by default, but this is not true. Some operations still need to be performed by external agents, and users must trust them to use the service. Thus, DeFi can be understood as a step further in the decentralisation philosophy.

Innovation is the key to achieving these services in a technical and economic sense. On the one hand, smart contract technology over blockchains (mainly Ethereum) and other platforms is a very novel technology that is still in the early stages of development. This technology congregates top-minded engineers working in a completely new field related to computer science, knowing that they have the huge task of improving the technology while maintaining compatibility, functionality, and with an immutable history of data and code releases. On the other hand, economic innovation redefines traditional economy in a completely new ecosystem, changing the sense of value, trust, coin, equity, or lending. The impact of economic innovation and specifically of DeFi on traditional economies, companies, and financial decision-makers will, in turn, bring more innovation in this field. The implications need to be carefully analysed.

Based on the experience with ICOs and their decline, countries' regulations have had an important role that now DeFi wants to surpass. As everything is on-chain and regulated through smart contracts, anonymity allows movement beyond the bounds of government regulations. DeFi may not be a success, but one of its main objectives is to be borderless. The obsession with smart contracts arose from the pursuit of transparency by DeFi. In this sense, as a smart contract is an open-source code, immutable when executed on a blockchain (even creators cannot change it), DeFi projects act as 'glass boxes' as everybody can see inside. DeFi is sometimes defined as a layered structure composed of settlement (e.g. Ethereum), asset (e.g. ERC20), protocol (e.g. Uniswap), application (e.g. Uniswap interface), and aggregation (e.g. Uniswap used by the third party) (Schär, 2020). To make it possible for all these elements to interact properly, considering the technical complexity of smart contracts, interoperability has become one objective to achieve in DeFi services.

These objectives, namely innovation, decentralisation, borderlessness, transparency, and interoperability (Chen and Bellavitis, 2020), act as a lever to push DeFi forward.

DeFi seems to be a big step toward the decentralisation of value, but it faces many challenges that could limit its expansion. These challenges, which are also blockchain challenges, should be addressed in this new crypto project. These include fraud, volatility, usability, regulatory uncertainty, privacy, building distributed trust, use of inflexible systems, lack of accountability, and lack of human presence (Chen and Bellavitis, 2020; Qin et al., 2020).

Safety problems have already appeared and it is easy to predict that they will continue to appear in the future. They can be in any part of the software; however, the magnitude of the security systems determines their impact on the ecosystem. During the ICO rush, some scams and bad practices appeared, and although DeFi is more transparent, technical and financial innovations could open doors to fraud, which can result in a "cost of dishonesty" (Domingo et al., 2020). Another important challenge relates to transparency, as it has a counterpart in possible privacy problems. If everything is recorded on public ledgers, there is anonymity but not necessarily privacy. All of this bears on the future of any project based on cryptocurrencies as it is always under the threat of government regulations, for example, by banning or constraining their use, similar to what happened to ICOs. Other challenges refer to the fact that decentralised financial services are consumed using digital applications, and it is common for a new software to lack usability, especially when dealing with very complex systems. In contrast, most DeFi cryptocurrencies are recent and have modest market caps, which can generate a lot of volatility when a sudden interest or disinterest occurs. Finally, the use of smart contracts ensures that it is not possible to change what is executed, even if both parties agree to a change. This means that human judgement does not count after a smart contract is released, which imposes important decision limitations. At this point, one question arises: if there is no central organisation responsible for the service, who is responsible for any error or failure? Fig. 1

2.5. DeFi as an investment opportunity

A good way to measure how DeFi projects are progressing is to examine the behaviour of its tradeable tokens. Tradeable tokens, often called cryptocurrencies, are used for governance, utility, security, or a combination of these for a given crypto project. Interest in a crypto project could lead a token to undergo variations in supply or demand and hence, in its price. Moreover, because cryptocurrencies are financial assets, they could be related to traditional assets as a significant part of their portfolio stocks (Gil-Alana et al., 2020). Some authors compared major cryptocurrencies and found evidence of integration in the prices of some altcoins, similar to what happens with stock indices (Gil-Alana et al., 2020). Other authors have created a value index of cryptocurrencies that persistently cross-correlates with the Dow-Jones Industrial Average (Zhang et al., 2018). Previous research compared the value of certain cryptocurrencies, such as Bitcoin and gold, reporting mixed results (Dyhrberg, 2016; Klein et al., 2018). Similarly, other studies suggest that there is a relationship between cryptocurrencies and broadly-defined commodity markets, such as energy, metals, and agriculture (Ji et al., 2019). More recently, a broader analysis based on the time and frequency domains found no relationship between the three largest cryptocurrencies and various other financial assets (Corbet et al., 2018).

Accordingly, more research is needed, and the following hypothesis is proposed.

H1. : Stock market volatility influences DeFi returns

2.6. Gold and oil in relation to crypto assets

The relationship between cryptocurrencies and gold has been and still is a very controversial topic. Some studies have demonstrated a correlation between Bitcoin and gold (Dyhrberg, 2016), but others (Klein et al., 2018) have not observed this correlation. One of the main ideas behind these studies is the possible use of Bitcoin as a gold-like asset, which is investment-safe. However, cryptocurrencies are beyond Bitcoin. Some studies include other cryptocurrencies, such as Ethereum

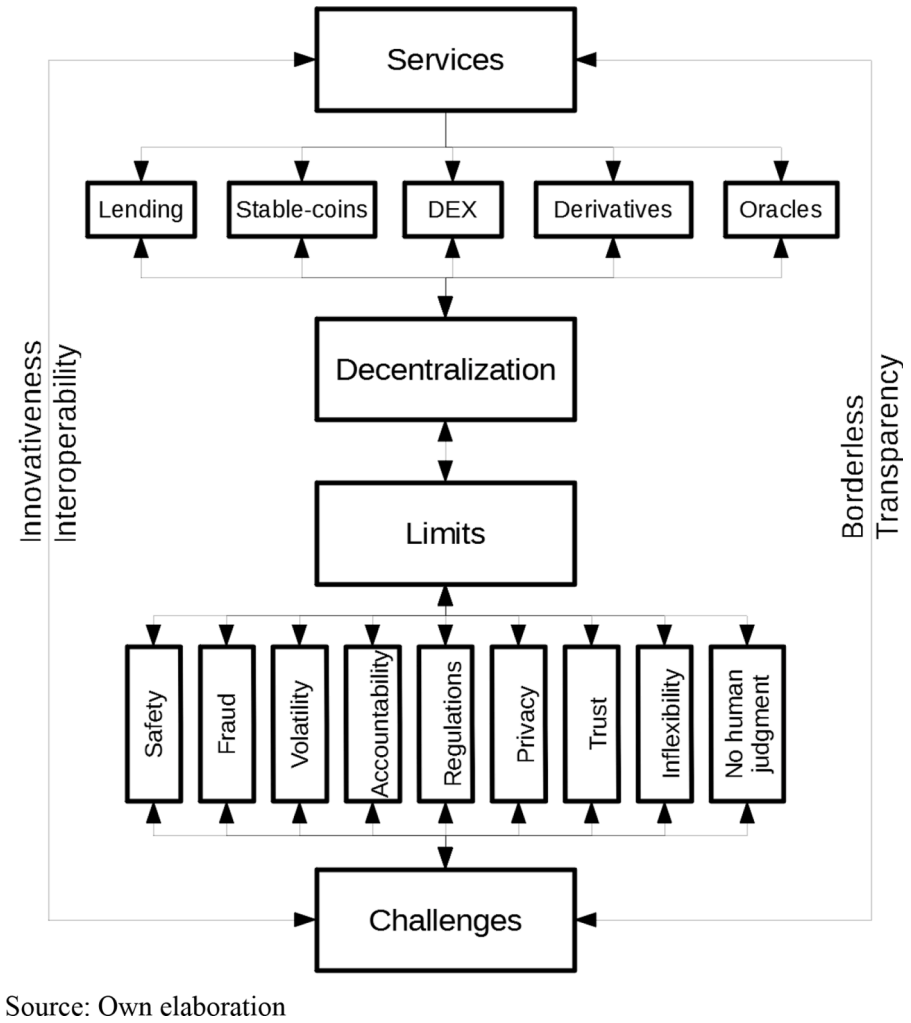


Fig. 1. DeFi's services, objectives, and challenges. Source: Own elaboration.

or Ripple (Beneki et al., 2019) but focus on volatility spillovers or bear markets (Kyriazis et al., 2019). Studies on other individual cryptocurrencies or class-grouped cryptocurrencies are lacking.

The same is true of the relationship between cryptocurrencies and oil. Al-Yahyaee et al. (2019) studied the relationship between the volatility of Bitcoin and oil markets, while Okorie and Lin (2020) focused on the volatility of some top cryptocurrencies and oil as a hedge for investors. Similarly, Yin et al. (2021) studied hedges, but in the particular case of oil market shocks. These studies showed a certain relationship between the volatility of oil and some cryptocurrencies, but the economic implications focused on hedges and portfolio management. We were unable to find studies that compared gold, oil assets, and DeFi tokens. Therefore, the following hypotheses are proposed:

H2. : DeFi return is influenced by energy sources such as oil (S&P GSCI crude oil index).

H3. : A DeFi is a safe asset, and its return is positively associated with the returns on other safe assets such as gold (S&P GSCI Gold Index).

2.7. Role of user-generated content in crypto assets

User-generated content (UGC) can be broadly materialised (e.g. wikis, evaluation websites, blogs, social media instant messaging apps) in different types of content formats (e.g. text, photo, videos, games) with countless purposes (e.g. entertainment, social-political campaigns, getting updated). Several researchers have analysed the diverse roles of UGC in the blockchain domain. Mai et al. (2018) used tweets and forum posts to forecast Bitcoin movement. Guske and Bendig (2018) found a relationship between the number of Twitter messages and the amount of funding raised. Rohr and Wright (2019) argued that the promoter or seller of a token often establishes communication centres (via slack, telegram, or message boards) to keep interested parties informed. Recently, Domingo et al. (2020) found that sentiment extracted from social networks positively influences ICO returns. To the best of our knowledge, no study has analysed the influence of UGC, namely, Telegram and Twitter, on DeFi returns. Hence, the following hypotheses are proposed:

H4. : Public user-generated content in Twitter influences DeFi returns.

H5. : Private user-generated content in Telegram influences DeFi returns.

2.8. Summary

As previously mentioned, the crypto landscape has changed over time, starting with Bitcoin, followed by smart contracts and initial coin offerings. Currently, DeFi is the main focus of change. Their novelty and limited evaluation of implications can generate a lot of volatility, mainly if certain events trigger a huge interest or disinterest and, consequently, huge price movements. Therefore, it is necessary to carefully analyse the relationship between DeFi volatility and stock market volatility measured through the Chicago Board Options Exchange Market

Volatility Index to better understand DeFi behaviour. The mixed results obtained by previous research and the lack of studies analysing the relationships between DeFi returns and stock market volatility or commodities such as gold or oil demand more scientific research.

3. Methodology

3.1. Data and variables

The database used in the analysis comprises the daily returns of DeFi, VIX, S&P GSCI crude oil index, S&P GSCI gold index, and the daily variation of DeFi mentions in Telegram chats and Twitter. The DeFi data, including social media mentions, were downloaded from Santiment.net. Santiment.net is a website and a crypto-based project that gathers data about cryptocurrencies – price, development activity, derivatives, or social media mentions, such as Twitter and Telegram. S&P GSCI Gold Index data were provided by the S&P Dow Jones Indices®. The VIX and S&P GSCI crude oil index data were downloaded from Investing.

Initially, we obtained data from 29 DeFi. We drop a DeFi from the sample if they have been unlisted for a long time. We also excluded DeFi, which only began in 2019 or 2020. Finally, we used a dataset consisting of 13 DeFi with returns from 1 July 2018 to 27 July 2020.

The daily returns of DeFi were calculated as:

$$Rn_{it} = (Pn_{it} - Pn_{it-1})/Pn_{it-1} \quad (1)$$

where Pn_{it} is the closing price of DeFi at moment t. This method was chosen because several DeFi have a closing price between 0 and 1, which makes it impossible to calculate returns following Campbell et al. (1997p.11).

However, the daily returns of the VIX, S&P GSCI crude oil index, and S&P GSCI gold index were calculated following Campbell et al. (1997p.11) method:

$$Ri_t = ln(Pi_t) - ln(Pi_{t-1}) \quad (2)$$

where Pit is the closing price of index i (VIX, S&P GSCI Crude Oil Index, or S&P GSCI Gold Index) at moment t. VIX, S&P GSCI Crude Oil Index, and S&P GSCI Gold Index have trading days without trade (Saturday and Sunday), but DeFi has trades in those days. Thus, we suppose that VIX, S&P GSCI Crude Oil Index, and S&P GSCI Gold Index have the same closing prices on days without trade as the last day with trade (Friday).

The daily variation of the mentions was calculated as follows:

$$TE_{it} = (TE_{it} - TE_{it-1})/TE_{it-1} \quad (3)$$

$$TW_{it} = (TW_{it} - TW_{it-1})/TW_{it-1} \quad (4)$$

where TE_{it} is the number of mentions in the Telegram chats of DeFi i at moment t, and TW_{it} is the number of mentions in Twitter of DeFi i at moment t.

Descriptive statistics for panel data with 9854 observations are presented in Table 1. The references in Telegram chats and Twitter have a lower number of observations because there are no mentions every day and because of the daily variation calculation. DeFi returns are represented by a dichotomous variable, where '1' is for positive or zero returns, and '0' for negative returns.

3.2. Model

To measure the relationship between assets in a nonlinear way, we use a logit-probit model, a widely used econometric method (Lennox, 1999). This nonlinear methodology has been applied to stocks and volatility by Thomas et al. (2014) and many others. We will take into consideration the results in the same way as other authors have done (Zelner, 2009).

This method allows us to test how financial market movements influence DeFi returns. The following logit-probit model was proposed:

$$Rn_{it} = \beta_0 + \beta_1 Rv_{it} + \beta_2 Rc_{it} + \beta_3 Rg_{it} + \beta_4 TE_{it} + \beta_5 TW_{it} + \varepsilon_t \quad (5)$$

where Rn_{it} represents the daily returns of DeFi (dichotomous variable), Rv_{it} is the daily return of VIX, Rc_{it} represents the daily returns of the S&P GSCI crude oil index, Rg_{it} is the daily returns of the S&P GSCI Gold Index, TE_{it} is the daily variation of mentions in Telegram chats, and TW_{it} represents the daily variation of mentions on Twitter.

Both probit and logit models were estimated using a population-averaged estimator. This means that the β measures the change in proportion to $y = 1$ for a unit increase in x (Neuhaus et al., 1991).

4. Results

Table 2 shows the correlations between the variables. The VIX and variation in Twitter mentions are negatively correlated with DeFi returns. This suggests that when volatility increases, DeFi returns decrease, and vice versa. In contrast, Twitter and Telegram chat references are positively correlated to DeFi returns. That is, an increase in Twitter or Telegram mentions could cause an increase in DeFi returns. Furthermore, the correlations between the S&P GSCI crude oil index returns and S&P GSCI gold index returns are positive, suggesting that an increase in these indices could cause an increase in DeFi returns. These suggest that DeFi movements are similar to market movements.

Table 3 shows the logit-probit results for the whole sample. The results were similar for both regressions, with coefficient estimates for all variables except the S&P GSCI Crude Oil Index returns and the daily variation of Twitter mentions are significant. DeFi reacts inversely to market volatility (VIX), moves in the same direction as gold, while the coefficient estimate of gold is significantly higher than the coefficient estimate of VIX. Results also suggest a strong correlation between DeFi and the references in Telegram.

Hausman's test showed that best fit was obtained with the probit model.

Table 2
Correlations.

	Rn	Rv	Rc	Rg	TE
Rv	−0.0700				
Rc	0.0237	−0.2400			
Rg	0.0439	0.0692	0.1126		
TE	0.0989	0.0107	0.0068	0.0055	
TW	0.0035	−0.0021	0.0108	0.0045	−0.0164

Table 1
Descriptive statistics.

Var	Obs.	Mean	Standard deviation	Min	Max	Skewness	Kurtosis
Rn	9854	0.4933022	0.4999805	0	1	0.0267936	1.000718
Rv	9854	0.0005676	0.0714943	−0.2662276	0.3917087	1.428916	9.379958
Rc	9854	−0.0015857	0.0394605	−0.5684967	0.2204784	−4.693289	71.98869
Rg	9854	0.0005395	0.008002	−0.047351	0.0560002	0.5418159	14.56932
TE	7755	0.4861318	3.026634	−1	125	17.2083	540.0146
TW	8001	0.0005696	0.0025155	−0.0854192	0.1071504	7.439103	628.839

Table 3
Logit-probit estimation results.

	Logit	Probit
Rv	−1.921994***	−1.204782***
Rc	0.0832564	0.0512301
Rg	10.18268***	6.401757***
TE	0.13784***	0.0762841***
TW	−5.48509	−3.927497
Const	−0.0776755***	−0.0476654***
Hausman	$chi^2(5) = 32.05$	(0.0000)

*** significance at 1% level; ** significance at 5% level; * significance at 10% level.

5. Discussion

The results suggest that DeFis can act as a safe haven, like gold. As shown by the positive correlation and significance of the logit-probit model estimates, parallelism with gold in terms of volatility is a hot topic in crypto research. Some authors such as Dyhrberg (2016) support it, while others such as Klein et al. (2018) do not. Both use GARCH methods, so this could be a methodology for future research. Indeed, our results are significant, but not in the higher range of significance, so other methodologies could help support hypothesis H3.

Following the seminal article by Whaley (2000), which outlined the methodology used in calculating VIX, Diavatopoulos et al., (2007) confirmed that implied volatility is superior to realized volatility in determining future returns. Interestingly, it has also been shown that the VIX is influenced by a range of characteristics demonstrated by non-technical investors in social media (López-Cabarcos et al., 2017). As implied volatility is our main interest, the VIX is an important variable to check if traditional stock volatility could be related to DeFi volatility. Our correlation and logit-probit model results support the inverse relationship, because the decrease in DeFi causes an increase in VIX, and vice versa. Our results are contrary to the findings of Lopez-Cabarcos et al. (2021) on a positive causality between VIX and returns on Bitcoin, and are more attuned to findings that VIX has a negative impact on risk premiums (Durand et al., 2011) or returns on stock indices (Giot, 2005). Here, we used the S&P GSCI Gold Index to track the COMEX gold futures. While it is widely tradeable, its close relationship with spot prices is not affected (Jena et al., 2018). The positive and statistically significant coefficient estimate of the gold proxy implies that Dyhrberg's (2016) claims that Bitcoin is positioned between gold, as a store value, and the US currency, and hence, can be put forward as a hedging instrument, could also be partially applied to DeFi. H1 and the more significant H3 seem to reinforce the idea of a safe haven for DeFi tokens.

The GSCI crude oil index showed no significant influence on the studied DeFi, indicating that H2 was not proven. Other studies comparing the influence between oil assets and Bitcoin have shown that the production of oil positively influences Bitcoin (Kaabia et al., 2020), or support the idea of Bitcoin as part of a diversified portfolio due to its low correlation with commodities, especially oil (Symitsi and Chalvatzis, 2019). The different time frames and methods make the comparison difficult.

The relationship between volatility and user-generated content shows that the daily variation of Telegram chats mentions influence DeFi returns positively (H4). Contrary to Pinero-Chousa et al. (2021), we find that the influence of Twitter mentions is insignificant (H5). These results are in line with other studies (López-Cabarcos et al., 2017), which show the influence of UGC and volatility depending on the type of investor and social network. In this study, Telegram chat groups of the DeFi project are more appealing to investors and people interested in the project. Contrastingly, Twitter is a more open platform for people with close interest in this type of project. As this is a subjective appreciation, more research can be conducted on the influence of UGC on DeFi.

As was mentioned, this study makes an important contribution to the extant body of knowledge by examining the impact of selected sentiment variables on DeFi returns. Telegram chats support our claim, contrary to some previous studies, whereas Twitter is not a potential tool for trading in DeFi. While commodity markets and stock market volatility are relevant in this study, the S&P 500 Index is not an important variable in determining DeFi returns.

6. Practical implications

DeFi can be analysed from three perspectives: DeFi as an alternative investment asset, the communication channels of DeFi, and the utility/limitations of DeFi.

In a very low or negative interest rate environment, investors are forced to take more risks or look for safe havens. Cryptocurrencies are a risky investment. However, this study found that gold influences DeFi, and VIX inversely influences DeFi. This makes DeFi a hedge against the volatility of the stock market and a safe haven. Consequently, these assets are useful in investment portfolios to balance risk and volatility.

We live in the information era, and everybody can write anything on any web or platform on the internet. Telegram is a private communication channel that usually focuses on one topic, whereas Twitter is a public and multi-topic channel. According to the results of this study, information from private and topic-focused platforms is more relevant for DeFi projects in terms of its correlation with token prices. As most crypto projects are present in both channels, this study helped assess their appropriateness.

The main objective of DeFi is to remove middlemen from financial services. Its adoption has exploded and it seems to be just starting. Depending on the implications of the investment and communication perspective, DeFi can be considered either an opportunity or a threat to the existing financial system.

7. Conclusion and future research

In terms of volatility, we analyse the influence of some traditional assets on some very specific crypto assets–decentralised finance tokens. As they are quite new, we thoroughly discuss these assets in the literature review. These results lead us to conclude that: (1) VIX inversely influences DeFi, (2) GSI crude oil has no influence on DeFi, and (3) the gold index influences DeFi, prompting us to claim that DeFi acts as a safe haven. This study can be added to a very broad and debated topic, and the alignment with methodologies and timeframes of other studies could be part of future research for better comparison.

The results obtained in user content generation are as follows: (4) Telegram chats influence DeFi and, (5) Twitter activity does not influence DeFi. As both assets and UGC platforms change significantly over time, the conclusion should be taken as specific to this case, requiring a broader study in terms of time, assets, and platforms.

As a general conclusion and taking into account that the technologies of DeFi assets are in their infancy, we can confirm there are relationships amongst some traditional assets, user-generated content, and DeFi. This influence may imply that DeFi tokens are safe havens, hedges for stock market volatility, and are linked to activity in Telegram's private groups.

The success of decentralised finance has attracted widespread interest from investors. Growth during the last months can be a sign for investors with a risk appetite looking for significant profit and a safe haven, or be included as high-risk investment instruments of a diversified portfolio. This should also be a warning for policymakers trying to avoid the situation that the ICO created several years ago. This situation of considerable returns from weak-value propositions leads to a rush of projects that lack solid foundations, or are even scams. As DeFi seems to have linked properties to some traditional investments and has the goal of recreating traditional finance services, we strongly recommend analysing it carefully.

DeFi is a novel product with limited dataset availability, and changes in the regulatory environment can have an impact on out-of-sample

results. Thus, the sample size is affected by the lack of longer time-series data. In this study, we include data from the pre-COVID 19 period, and the results could vary if analysed during and after the current health crisis. With an increase in the number of DeFi, a larger sample size is possible in the future and illiquidity issues can be addressed more meticulously. It is also advisable to include the variations in current sentiment variables and examine the introduction of new ones with the advancement of AI. In future research, it will be necessary to examine not only the COVID-19 period, but also the post-pandemic period.

In summary, the DeFi landscape of projects and their behaviour as a category of crypto assets is rising and may change significantly in the future. This demands much more research effort.

CRediT authorship contribution statement

Juan Piñeiro-Chousa: Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Supervision. **M. Ángeles López-Cabarcos:** Conceptualization, Methodology, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Supervision. **Aleksandar Sevic:** Conceptualization, Methodology, Formal analysis, Investigation, Resources, Data curation, Writing – original draft, Writing – review & editing. **Isaac González-López:** Conceptualization, Methodology, Formal analysis, Data curation, Investigation, Writing – original draft, Writing – review & editing.

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